



S.B. JAIN INSTITUTE OF TECHNOLOGY MANAGEMENT & RESEARCH, NAGPUR

Practical 08

Aim: A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Least recently used page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

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Academic Session:

Date of Performance:

Date of Submission:

CODE :-

```
M ~  
Shweta Bambode@Shweta MSYS ~  
$ nano lru.c
```

```
M ~  
GNU nano 8.7 lru.c  
#include <stdio.h>  
  
int main() {  
    int frames[3];  
    int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};  
    int time[3];  
    int i, j, k;  
    int hit = 0, fault = 0;  
    int n = 12, frames_count = 3;  
    int counter = 0, flag;  
  
    // initialize frames  
    for(i = 0; i < frames_count; i++) {  
        frames[i] = -1;  
    }  
  
    for(i = 0; i < n; i++) {  
        flag = 0;  
  
        // check page hit  
        for(j = 0; j < frames_count; j++) {  
            if(frames[j] == pages[i]) {  
                hit++;  
                counter++;  
                time[j] = counter;  
                flag = 1;  
                break;  
            }  
        }  
    }  
}
```

```

GNU nano 8.7                                     lru.c
    }
}

// page fault
if(flag == 0) {
    int min = 0;

    for(j = 1; j < frames_count; j++) {
        if(time[j] < time[min])
            min = j;
    }

    frames[min] = pages[i];
    counter++;
    time[min] = counter;
    fault++;
}

}

float hit_percent = (hit * 100.0) / n;
float fault_percent = (fault * 100.0) / n;

printf("Total Page Hits: %d\n", hit);
printf("Total Page Faults: %d\n", fault);
printf("Page Hit Percentage: %.2f%%\n", hit_percent);
printf("Page Fault Percentage: %.2f%%\n", fault_percent);

return 0;
}

```

OUTPUT:-

```

Shweta Bamcode@Shweta MSYS ~
$ gcc lru.c -o lru

Shweta Bamcode@Shweta MSYS ~
$ ./lru
Total Page Hits: 2
Total Page Faults: 10
Page Hit Percentage: 16.67%
Page Fault Percentage: 83.33%

```

